1 Solve the simultaneous equations

$$4x + 3y = 18$$
$$x - 3y = 7$$

$5x = 25$

$x = 5$

$4(5) + 3y = 18$

$20 + 3y = 18$

$3y = -2$

$y = \frac{-2}{3}$

$x =$ 5

$y =$ $-2/3$

(Total for question 1 is 3 marks)

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2 Solve the simultaneous equations

$$x - 3y = -23 \qquad \times 5$$
$$5x + 2y = 4$$

$5x - 15y = -115$

$5x + 2y = 4$

$-17y = -119$

$y = \frac{119}{17}$

$= 7$

$5x + 2(7) = 4$

$5x + 14 = 4$

$5x = -10$

$x = -2$

$x =$ -2

$y =$ 7

(Total for question 2 is 3 marks)

3 Solve the simultaneous equations

$$2x + 5y = -10$$
$$2x - y = 8$$

$6y = -18$

$y = -3$

$2x + 5(-3) = -10$

$2x - 15 = -10$

$2x = 5$

$x = \frac{5}{2}$

$x =$ $\frac{5}{2}$

$y =$ -3

(Total for question 3 is 3 marks)

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4 Solve the simultaneous equations

$$4x + 2y = 10 \quad \times 5$$
$$5x + 3y = 12 \quad \times 4$$

$20x + 10y = 50$

$20x + 12y = 48$

$-2y = 2$

$y = -1$

$4x + 2(-1) = 10$

$4x - 2 = 10$

$4x = 12$

$x = 3$

$x =$ 3

$y =$ -1

(Total for question 4 is 3 marks)

5 Solve the simultaneous equations

$$2x + 5y = 4$$
$$7x - 5y = -1$$

$9x = 3$

$x = \frac{3}{9} = \frac{1}{3}$

$2\left(\frac{1}{3}\right) + 5y = 4$

$\frac{2}{3} + 5y = 4$

$\frac{2}{3} + 5y = \frac{12}{3}$

$5y = \frac{10}{3}$

$y = \frac{2}{3}$

$x =$ $\frac{1}{3}$

$y =$ $\frac{2}{3}$

(Total for question 5 is 3 marks)

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6 Solve the simultaneous equations

$$3x - 2y = 7$$
$$7x + 2y = 13$$

$10x = 20$

$x = 2$

$7(2) + 2y = 13$

$14 + 2y = 13$

$2y = -1$

$y = -\frac{1}{2}$

$x = 2$ $y = -\frac{1}{2}$

(Total for question 6 is 3 marks)

7 Solve the simultaneous equations

$$2x - 3y = 4 \qquad \times 2$$
$$4x - y = 13$$

$$4x - 6y = 8$$
$$4x - y = 13$$

$$-5y = -5$$
$$y = 1$$

$$4x - 1 = 13$$
$$4x = 14$$
$$x = \frac{14}{4} = \frac{7}{2}$$

$x =$ $\frac{7}{2}$

$y =$ 1

(Total for question 8 is 3 marks)

---

8 Solve the simultaneous equations

$$3x + y = 15 \qquad \times 2$$
$$5x + 2y = 24$$

$$6x + 2y = 30$$
$$5x + 2y = 24$$
$$x = 6$$

$$3(6) + y = 15$$
$$18 + y = 15$$
$$y = -3$$

$x = 6 \quad y = -3$

(Total for question 8 is 3 marks)

9 Solve the simultaneous equations

$$3x - y = -4 \qquad \times 2$$
$$2x - 3y = 9 \qquad \times 3$$

$$6x - 2y = -8$$
$$6x - 9y = 27$$

$$7y = -35$$
$$y = -5$$

$$3x - (-5) = -4$$
$$3x + 5 = -4$$
$$3x = -9$$
$$x = -3$$

$x =$ ..... -3 .....

$y =$ ..... -5 .....

(Total for question 9 is 3 marks)

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10 Solve the simultaneous equations

$$6x + 5y = 4.5$$
$$3x - 2y = 9 \qquad \times 2$$

$$6x - 4y = 18$$
$$6x + 5y = 4.5$$

$$-9y = 13.5$$
$$y = -\frac{13.5}{9} = \frac{-27}{18} = -\frac{3}{2}$$

$$3x - 2\left(\frac{-3}{2}\right) = 9$$

$$3x + 3 = 9$$

$$3x = \cancel{12}\ 6$$
$$x = \cancel{4}\ 2$$

$x =$ ..... $\cancel{4}\ 2$ .....

$y =$ ..... $-\frac{3}{2}$ .....

(Total for question 10 is 3 marks)

11 Solve the simultaneous equations

$$3x = 9 + y$$
$$x + 5y = 5$$

$3x - y = 9$

$x + 5y = 5 \quad \times 3$

$3x - y = 9$

$3x + 15y = 15$

$-16y = -6$

$y = \frac{6}{16} = \frac{3}{8}$

$x + 5\left(\frac{3}{8}\right) = 5$

$x + \frac{15}{8} = 5$

$x = \frac{40}{8} - \frac{15}{8} = \frac{25}{8}$

$x =$ $\frac{25}{8}$

$y =$ $\frac{3}{8}$

(Total for question 11 is 3 marks)

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12 Solve the simultaneous equations

$$3y + 11 = 4x$$
$$10x + 2y + 1 = 0$$

$\cancel{3y}$ $11 = 4x - 3y$

$4x - 3y = 11$ (1) $\times 5$

$10x + 2y = -1$ (2) $\times 2$

$20x - 15y = 55$

$20x + 4y = -2$

$-19y = 57$

$y = -3$

$4x - 3(-3) = 11$

$4x + 9 = 11$

$4x = 2$

$x = \frac{1}{2}$

$x =$ $\frac{1}{2}$

$y =$ -3

(Total for question 12 is 3 marks)

13 In a shop 2 coffees and 3 cakes cost £9.95
In the same shop 1 coffee and 4 cakes cost £10.35.
Work out the price for one coffee and the price for one cake.

x y x y

$2x + 3y = 9.95$

$x + 4y = 10.35 \quad \times 2$

$2x + 8y = 20.70$

$-\quad -\quad -$

$2x + 3y = 9.95$

$5y = 10.75$

$y = 2.15$

$x + 4(2.15) = 10.35$

$x + 8.60 = 10.35$

$x = 1.75$

Coffee £ 1.75

Cake £ 2.15

(Total for question 13 is 3 marks)

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14 Sweets are sold in small packs and in big packs.
There is a total of 175 sweets in 4 small packs and 3 big packs.
There is a total of 154 sweets in 5 small packs and 2 big packs.
Work out the number of sweets in each small pack and in each big pack.

$4s + 3b = 175 \quad \times 2$

$5s + 2b = 154 \quad \times 3$

$8s + 6b = 350$

$-\quad -\quad -$

$15s + 6b = 462$

$-7s = -112$

$s = 16$

$5(16) + 2b = 154$

$80 + 2b = 154$

$2b = 74$

$b = 37$

Small Pack 16

Big Pack 37

(Total for question 14 is 3 marks)